# AGHA MALIK BAIG MIRZA

Computer Engineer
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## SUMMARY

M.S. Computer Engineering student at CSUN, graduating December 2026, seeking full-time RTL design, design verification, ASIC/SoC and FPGA roles. Built UVM testbenches for AMBA APB and AXI4 memory slaves in coursework and personal projects. Strong in Verilog and hands-on with SystemVerilog.

## TECHNICAL SKILLS

- **Verification:** UVM, SVA, constrained-random verification, functional coverage, directed testbenches.
- **HDLs and programming:** Verilog, SystemVerilog, RTL design and debugging, Python, C.
- **Tools:** Linux command line, shell scripts, Synopsys VCS and DVE, QuestaSim/ModelSim, Xilinx Vivado, GTKWave, EDA Playground.
- **Protocols:** AMBA APB, AXI4, SPI, I2C, UART, synchronous FIFO.

## ACADEMIC PROJECT

**Design and UVM Verification of AXI4 Memory | github.com/mirzamalikbaig/axi4_uvm**

- Wrote a UVM environment for a memory-mapped AXI4 slave covering the AW, W, B, AR and R channels.
- Verified VALID/READY handshakes, independent AW/W timing, backpressure, WSTRB, responses, reset behavior and read-after-write data integrity.
- Added scoreboard-based memory checking and functional coverage for transactions and handshakes.
- Ran VCS simulation and coverage from shell scripts: 98.15% functional coverage, 0 UVM errors, 12 directed scenarios.

## PERSONAL PROJECTS

**Design and UVM Verification of APB Memory**

- Designed and verified an AMBA APB memory slave using UVM with constrained-random stimulus, a scoreboard, assertions and functional coverage.
- Wrote self-checking tests for reset, read and write transfers, boundary values, randomized address/data and write-then-read integrity.
- Debugged protocol and data mismatches with scoreboard failures and VCS waveforms, fixing RTL and testbench issues before coverage closure.
- Achieved 100% functional, 100% line and branch, 70.43% toggle and 60% condition coverage.

**SystemVerilog FIFO Verification Environment**

- Developed a class-based verification environment for a parameterized synchronous FIFO using transaction, generator, driver, monitor and scoreboard components.
- Verified read/write, simultaneous operations, full/empty, overflow/underflow and boundary conditions using directed and randomized tests, coverage, assertions and waveform analysis.

## EXPERIENCE

**Verification Intern, Aisemi Pvt. Ltd. | Jan 2024 – Apr 2024**

- Four-month internship on SystemVerilog, RTL verification fundamentals and semiconductor concepts.
- Developed and simulated SystemVerilog RTL and testbenches for combinational and sequential logic.
- Ran Synopsys simulations, inspected waveforms and debugged functional mismatches.

**Industrial Trainee, Electronics Corporation of India Limited (ECIL) | Nov 2019 – Mar 2020**

- Industrial training at ECIL, a Government of India enterprise, with exposure to electronic systems, digital circuits and laboratory practices.
- Performed supervised circuit testing, signal measurement and basic hardware troubleshooting.

## EDUCATION

**M.S. in Computer Engineering, California State University, Northridge (CSUN) | Aug 2024 – Dec 2026**

Coursework: Verilog & SystemVerilog, ASIC Development, System-on-Chip Design, Computer Systems Architecture, Computer Arithmetic Design.

**B.E. in Electronics & Communication Engineering, Osmania University | Jul 2021 – Jun 2023**

Coursework: Digital Logic Design, VLSI Design, Microprocessors & Microcontrollers, Embedded Systems.